

# Machine Learning

## Supervised Learning: Linear Regression



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Machine Learning (25SC2107E)

## Session Outcome

- Understand linear regression as a predictive model and explain how parameters are learned via *normal equations* and *gradient descent*.

# Regression

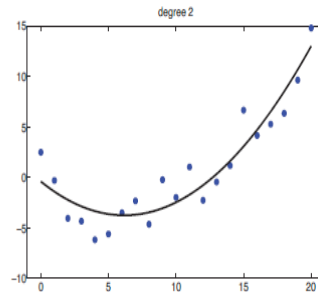
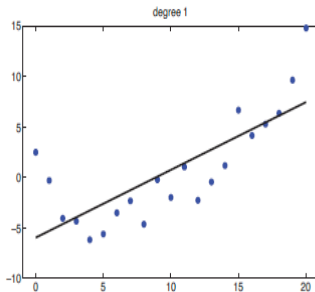
# Regression

**Regression** is a supervised learning technique used to predict continuous numerical values by learning relationships between input variables (features) and an output variable (target).

## Types

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Stepwise Regression
- Ridge Regression
- Lasso Regression
- ElasticNet Regression

# Linear vs Polynomial



Source: Murphy

Figure: Linear regression vs Polynomial (degree-2) regression

# Linear Regression

**Linear Regression** is a supervised machine learning algorithm used to predict continuous numerical values by finding a straight-line relationship between an input feature and a target output.

- *Inputs*: Independent variables, predictors, features
- *Outputs*: Dependent variables, responses

Linear regression model prediction is given by:

$$\hat{y} = h_{\theta}(x) = \theta_0 + \theta_1 x_1 + \cdots + \theta_d x_d = \theta^T \mathbf{x}$$

where

- $\hat{y}$  is the predicted values
- The number of features= $d + 1$
- $x_i$  is the  $i^{th}$  feature value
- $\theta_j$  is the  $j^{th}$  model parameter

# Linear regression- Vector Notation

The Linear regression model is given by

$$\hat{y} = h_{\theta}(\mathbf{x}) = \theta \cdot \mathbf{x} = \theta^T \mathbf{x}$$

where

- $h_{\theta}$  is the *hypothesis* function
- In linear regression, the **hypothesis function** is the equation used to predict the target output ( $y$ ) from the given input features ( $x$ ).
- $\theta$  is the model's *parameter* vector  $\theta = [\theta_0, \theta_1, \dots, \theta_d]^T$
- $\mathbf{x}$  is the *feature* vector  $\mathbf{x} = [1, x_1, \dots, x_d]^T$

Question: Given data for  $m$  observations, how to train the model? Find the parameter vector  $\theta$  so that the model best fits the training set

# Example

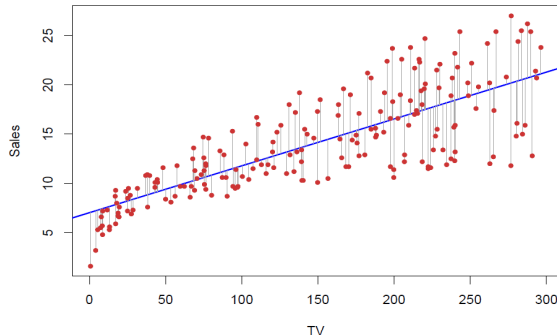
Consider the Sales of a product depends on TV advertisements

$$\text{Sales} \approx \theta_0 + \theta_1 \text{TV} \quad (1)$$

$$y = \theta_0 + \theta_1 x \quad (2)$$

Problem: Given  $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$  find  $\theta_0, \theta_1$   
minimize mean square error

$$\text{MSE} = \frac{1}{n} ((y_1 - \theta_0 - \theta_1 x_1)^2 + (y_2 - \theta_0 - \theta_1 x_2)^2 + \dots + (y_n - \theta_0 - \theta_1 x_n)^2)$$



Source: James

**Figure:** The fit/model is found by minimizing the distance



# Mean Squared Error (MSE)

The MSE of a linear regression hypothesis  $h_\theta$  on a training set  $\mathbf{X}$  is given by

$$MSE(\mathbf{X}, h_\theta) = J = \frac{1}{m} \sum_{i=1}^m (\theta^T \mathbf{x}^{(i)} - y^{(i)})^2 = \frac{1}{m} (\mathbf{X}\theta - \mathbf{y})^T (\mathbf{X}\theta - \mathbf{y})$$

Expand MSE  $J = \frac{1}{m} (\theta^T \mathbf{X}^T \mathbf{X} \theta - 2 \theta^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y})$ .

Take the gradient w.r.t.  $\theta$  (matrix calculus:  $\nabla_\theta \theta^T A \theta = 2A\theta$ ,  $\nabla_\theta \theta^T b = b$ ):

$$\nabla_\theta J = \frac{1}{n} (2 \mathbf{X}^T \mathbf{X} \theta - 2 \mathbf{X}^T \mathbf{y})$$

Set the gradient to zero and solve for  $\hat{\theta}$ . We get the **Normal Equation**

$$\hat{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\mathbf{X} = \begin{bmatrix} 1 & x_0^1 & \cdots & x_d^1 \\ 1 & x_0^2 & \cdots & x_d^2 \\ \vdots & & & \\ 1 & x_0^m & & x_d^m \end{bmatrix}$$

$$\mathbf{y} = \begin{bmatrix} y^1 \\ y^2 \\ \vdots \\ y^m \end{bmatrix}$$

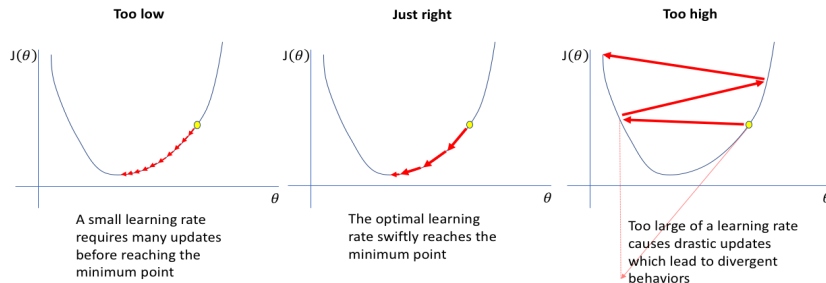
# Computational complexity in Matrix inversion

- The Normal equation computes the inverse of  $X^T X$ , which is an  $(d + 1) \times (d + 1)$  matrix (where  $d$  is the number of features).
- The computational complexity of inverting such a matrix is typically about  $O(d^3)$ .
- When the number of features are large (100,000) matrix inversions required memory and time.
- So use *gradient descent* to find the parameter vector  $\hat{\theta}$

# Gradient Descent

$$\theta^{(i+1)} = \theta^{(i)} - \eta \nabla_{\theta} J(\theta^{(t)}), \quad \nabla_{\theta} J = \frac{2}{n} \mathbf{X}^T (\mathbf{X}\theta - \mathbf{y})$$

where  $\eta$  is the learning rate



Source: Duchesnay

## Example 1 (Simple Linear Regression)

**Problem:** Consider five weeks' sales data. Apply linear regression to predict the 7<sup>th</sup> and 12<sup>th</sup> week sales.

| $x_i$ (Week) | $y_j$ (Sales in Thousands) |
|--------------|----------------------------|
| 1            | 1.2                        |
| 2            | 1.8                        |
| 3            | 2.6                        |
| 4            | 3.2                        |
| 5            | 3.8                        |

**Linear regression equation:**  $y = \theta + \theta_1 * x$

## Example 1 (Simple Linear Regression)

Define the Matrix  $\mathbf{X}$  and Target Vector  $\mathbf{y}$ :

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{pmatrix} \quad \text{and} \quad \mathbf{y} = \begin{pmatrix} 1.2 \\ 1.8 \\ 2.6 \\ 3.2 \\ 3.8 \end{pmatrix}$$

$$\mathbf{X}^T \mathbf{X} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}$$

$$(\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} = \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix}$$

$$\mathbf{X}^T \mathbf{Y} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1.2 \\ 1.8 \\ 2.6 \\ 3.2 \\ 3.8 \end{pmatrix} = \begin{pmatrix} 12.6 \\ 44.4 \end{pmatrix}$$

Solve for  $\hat{\theta}$ :

$$\hat{\theta} = (X^T X)^{-1} X^T Y = \begin{pmatrix} 1.1 & -0.3 \\ -0.3 & 0.1 \end{pmatrix} \begin{pmatrix} 12.6 \\ 44.4 \end{pmatrix}$$

$$\hat{\theta} = \begin{pmatrix} 1.1(12.6) - 0.3(44.4) \\ -0.3(12.6) + 0.1(44.4) \end{pmatrix} = \begin{pmatrix} 13.86 - 13.32 \\ -3.78 + 4.44 \end{pmatrix} = \begin{pmatrix} 0.54 \\ 0.66 \end{pmatrix}$$

**Constructed Model:**

- Intercept ( $\theta_0$ ) = 0.54
- Slope ( $\theta_1$ ) = 0.66

$$y = 0.54 + 0.66 * x$$

**Predictions:**

- The predicted 7<sup>th</sup> week sale (when  $x = 7$ ) is:

$$y = 0.54 + 0.66 \times 7 = 5.16$$

- The predicted 12<sup>th</sup> week sale (when  $x = 12$ ) is:

$$y = 0.54 + 0.66 \times 12 = 8.46$$

## Example 2 (Multiple Linear Regression)

**Problem:** Find the multiple regression equation for the given product sales data.

| $x_1$ (Product 1 Sales) | $x_2$ (Product 2 Sales) | $Y$ (Weekly Sales) |
|-------------------------|-------------------------|--------------------|
| 1                       | 4                       | 1                  |
| 2                       | 5                       | 6                  |
| 3                       | 8                       | 8                  |
| 4                       | 2                       | 12                 |

**Regression Equation:**  $y = \theta_0 + \theta_1 x_1 + \theta_2 x_2$

**Coefficient Formula:**  $\hat{\theta} = ((X^T X)^{-1} X^T) Y$

## Example 2

**Define Matrices  $X$  and  $Y$ :**

$$X = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 2 & 5 \\ 1 & 3 & 8 \\ 1 & 4 & 2 \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} 1 \\ 6 \\ 8 \\ 12 \end{pmatrix}$$

**Calculate  $X^T X$  and its inverse:**

$$X^T X = \begin{pmatrix} 4 & 10 & 19 \\ 10 & 30 & 46 \\ 19 & 46 & 109 \end{pmatrix}$$

$$(X^T X)^{-1} = \begin{pmatrix} 3.15 & -0.59 & -0.30 \\ -0.59 & 0.20 & 0.016 \\ -0.30 & 0.016 & 0.054 \end{pmatrix}$$



## Example 2

**Calculate Coefficients ( $\hat{a}$ ):**

$$\hat{a} = ((X^T X)^{-1} X^T) Y = \begin{pmatrix} -1.69 \\ 3.48 \\ -0.05 \end{pmatrix}$$

**Extracted Values:**

- Intercept ( $\theta_0$ ) = -1.69
- Coefficient for  $x_1$  ( $\theta_1$ ) = 3.48
- Coefficient for  $x_2$  ( $\theta_2$ ) = -0.05

**Constructed Model:**

$$y = -1.69 + 3.48x_1 - 0.05x_2$$

# MCQs

- ❶ In  $\hat{y} = \theta_0 + \theta_1 x_1$ , the parameter  $\theta_1$  represents  
(a) the intercept (b) change in  $\hat{y}$  per unit of  $x_1$  (c) the residual (d) the learning rate **Answer: (b)**
- ❷ The MSE cost surface for linear regression is  
(a) non-convex with many minima (b) convex with one global minimum (c) flat (d) discontinuous  
**Answer: (b)**
- ❸ The normal-equation solution is  
(a)  $(\mathbf{X}\mathbf{X}^\top)^{-1}\mathbf{y}$  (b)  $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$  (c)  $\mathbf{X}^{-1}\mathbf{y}$  (d)  $\mathbf{X}^\top\mathbf{y}$  **Answer: (b)**
- ❹ Predicting salary package in LPA from student features is  
(a) classification (b) regression (c) clustering (d) ranking **Answer: (b)**

# MCQs

- ❶ Squaring residuals in MSE means  
(a) sign is ignored and large errors dominate    (b) small errors dominate    (c) outliers are ignored    (d) cost is non-differentiable  
**Answer: (a)**
- ❷ The normal equations fail when  $\mathbf{X}^T \mathbf{X}$  is singular, e.g. when  
(a)  $n$  is large    (b) two features are perfectly collinear    (c)  $y$  is scaled    (d)  $\eta$  is small  
**Answer: (b)**
- ❸ A learning rate that is too large causes gradient descent to  
(a) converge slowly    (b) diverge/oscillate    (c) stop at the optimum    (d) need fewer epochs  
**Answer: (b)**
- ❹ For  $d = 100,000$  features, the practical trainer is  
(a) matrix inversion    (b) gradient descent    (c) Cramer's rule    (d) exhaustive search  
**Answer: (b)**